

Worksheet 102 · Spot on facts, pp. 226–227

# Technologies that change society

The key facts at a glance — the connections are yours to make. Your basis for the Kolloquium.

## 1 Developments in the 21st century

### Genetic engineering

**CRISPR** edits the genetic code precisely.

+ drought-resistant crops, new treatments

- morality; *interference* with DNA

### Eco-technology

Solar, hydrogen and wind power are cheaper; more e-cars.

+ less fossil fuel, ocean clean-ups

- raw materials (lithium) harm the environment

### AI & digital content

Prompts to large language models create texts, media, programs.

+ information re-used in myriad contexts

- where does content come from — can we trust it?

**Connect:** All three bring a hope and a new problem. Which development shows this most clearly — and why?

*E.g. eco-technology: e-cars reduce the use of fossil fuels, but the lithium for their batteries creates new environmental problems — one solution produces the next problem.*

## 2 Artificial intelligence (AI)

AI = the simulation of human intelligence in machines: learning, problem-solving, reasoning, perceiving.

### Weak / narrow AI

single tasks, e.g. personal assistants, video games

### Strong / general AI

complex, human-like, e.g. (book) self-driving cars, robots\*

### Impact

+ everyday life, education, health

- jobs, deepfakes, misinformation, copyright

\*Most experts today call self-driving cars *narrow AI*; *general AI* means a still hypothetical human-level intelligence.

**Connect:** Why does checking data matter more than ever now that AI creates content?

*AI can produce convincing texts and deepfakes, and it is often impossible to tell where content comes from. Checking sources is the only protection against manipulation and misinformation.*

## 3 Augmented reality (AR) and virtual reality (VR) — both extended reality (XR)

### AR

What it does adds digital overlays to the real world

Devices phone, tablet (camera)

Used in entertainment, medicine, shopping

### VR

replaces the real world with a 360° environment

headset, motion controllers

gaming, tourism, education, medicine, warfare

**Connect:** How are AR, VR and AI related? Name one example you know (e.g. Virtual Helsinki, p. 230).

*AR and VR are both types of XR; AI can make them smarter and more realistic (e.g. AI avatars, generated worlds). Example: Virtual Helsinki is VR, a digital twin of the city.*

## 4 Utopia or dystopia — where will technology take us?

**Utopia** ou-topos = “no place”, eu-topos = “good place” (Thomas More, 1516).

An **ideal society** that criticises the present by showing a better world.

e.g. Butler, *Erewhon* (1872); Bellamy, *Looking Backward* (1888)

**Dystopia** dys-topos = “bad place”.

A **worst-case society**, often in the near future, that warns about *contemporary* problems.

e.g. *Brave New World* (1932), *Nineteen Eighty-Four* (1949), *The Hunger Games* (2008)

**Connect:** Which technology from boxes 1–3 could be the starting point of a dystopia? Describe that society in one or two sentences.

*E.g. AI + VR: people spend most of their lives in virtual worlds whose content is selected by AI. Reality loses importance and opinions can be steered without anyone noticing — a society like Brave New World, kept happy and passive by technology.*

# Kolloquium Carousel

In Part 2 of the Kolloquium you answer questions on topics from other semesters — no notes. Train it now.

## How it works

- 1 Double circle: inside **candidates**, outside **examiners**.
- 2 Examiner reads a card — 15 s to think, **2 min** to answer.
- 3 Follow-up question (1 min), then rubric + two stars and a wish.
- 4 Outer circle moves on. Swap roles after two rounds.

## Answer like a candidate

Define



Explain



Example



Evaluate

## Question cards (follow-up in italics)

### 1 AR vs VR

Explain the difference between AR and VR.

*Follow-up: Which will change everyday life more?*

### 2 AI

What is AI? Weak vs strong AI.

*Follow-up: Where does AI help — where does it worry you?*

### 3 Trust

Why is it harder to know where online content comes from?

*Follow-up: How can you protect yourself?*

### 4 Genes

What does CRISPR make possible?

*Follow-up: Where would you draw the line?*

### 5 Terms

Where do *utopia* and *dystopia* come from?

*Follow-up: Why are dystopias so popular today?*

### 6 Big picture

Which technology is the most life-changing?

*Follow-up: How will it affect your children's lives?*

## Examiner's sheet 0 = not yet · 1 = partly · 2 = mostly · 3 = fully

|                                                | Round 1 | Round 2 |
|------------------------------------------------|---------|---------|
| Candidate                                      |         |         |
| <b>Content</b> — accurate facts and terms      | /3      | /3      |
| <b>Structure</b> — clear line of thought       | /3      | /3      |
| <b>Language</b> — topic words, range, accuracy | /3      | /3      |
| <b>Interaction</b> — reacts to the follow-up   | /3      | /3      |
| <b>Total</b>                                   | /12     | /12     |

**After the carousel — your Kolloquium note** Which card was hardest? What would you need to answer it well next time?

*Individual answers, e.g. card 4 (Genes): I need the exact definition of CRISPR, one concrete example (drought-resistant crops) and a clear position on where to draw the line (curing diseases yes, "designer babies" no).*